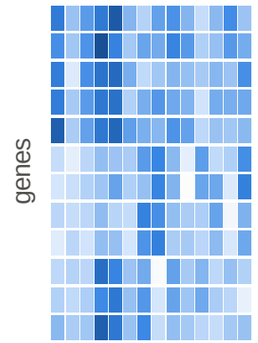
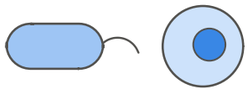
Adding a softmax and a multi-step walk to CLR's normalization recovers large *E. coli* regulons better (16% over CLR), but loses resolution on tight protein complexes

CLR (Faith et al. 2007)

1 Expression compendium



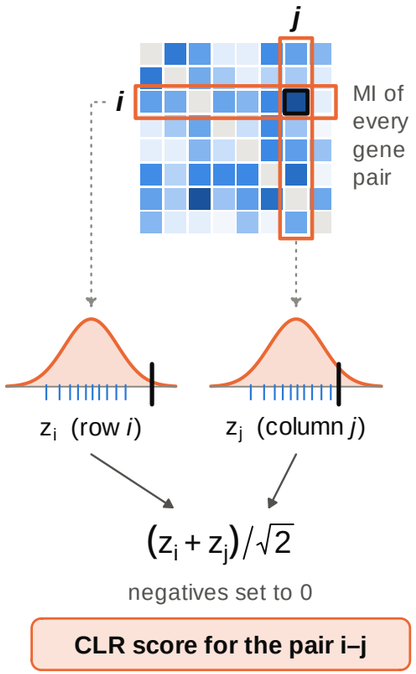


E. coli human

M3D, EcoMAC microarrays — K562 Perturb-seq

2 Mutual information, z-scored against each gene's own background

existing method

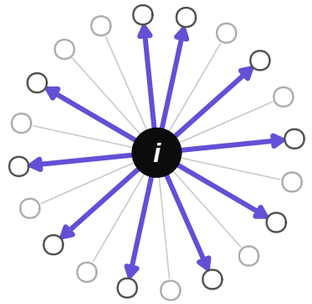


CLR score for the pair i–j

Published CLR stops here: a score for each gene pair

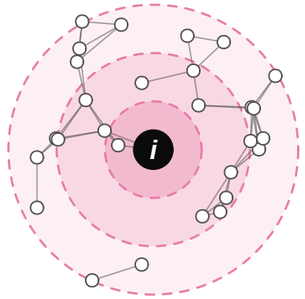
This work: steps taken from attention

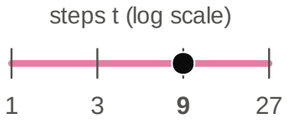
3 Softmax with per-gene temperature



~10 effective partners per gene, set in advance

4 Multi-step walk, number of steps chosen without labels





t = 9: chosen on held-out labs

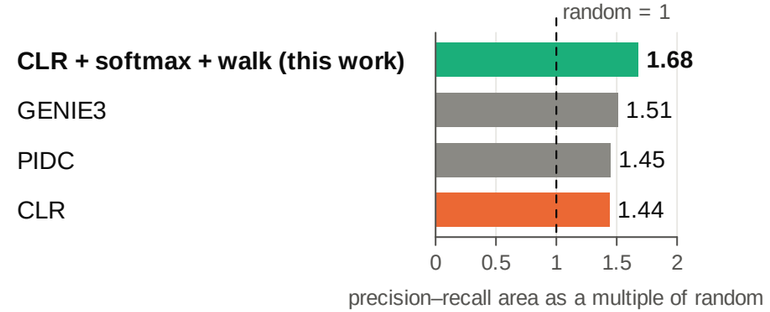
No learned parameters

Transformer analogy: MI ≈ query·key score; CLR z-score ≈ scaled logit; softmax and walk ≈ attention and stacked layers

5 What it recovers

E. coli regulons (experimentally supported, RegulonDB)

regulon = genes controlled by one transcription factor (TF)



per regulon: tie with CLR and GENIE3; ahead in 1,000/1,000 resamplings vs CLR

Protein complexes (K562 CORUM; mostly ribosome): this work loses 1.8×

complex = proteins that bind as one machine

